



同濟大學
TONGJI UNIVERSITY

 ADMIS Lab
先进数据与机器智能系统实验室

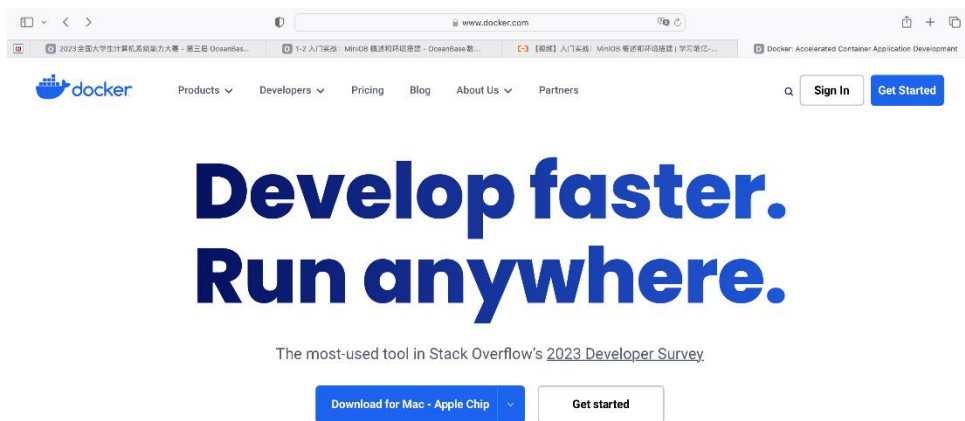
miniOB 环境搭建指导

基于Docker

2025年3月20日

Docker安装

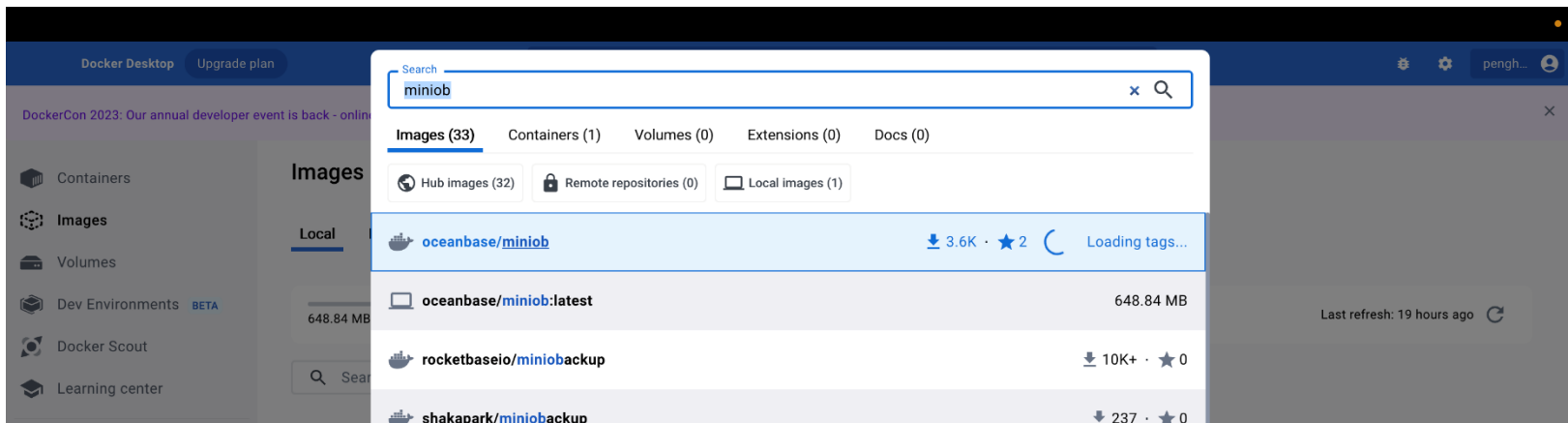
- 为了便于后续实验，统一使用Docker搭建数据库环境。
 - 前往 <https://www.docker.com> 网站下载电脑对应版本的安装包



- 安装后，按照要求注册/登录账号

下载镜像

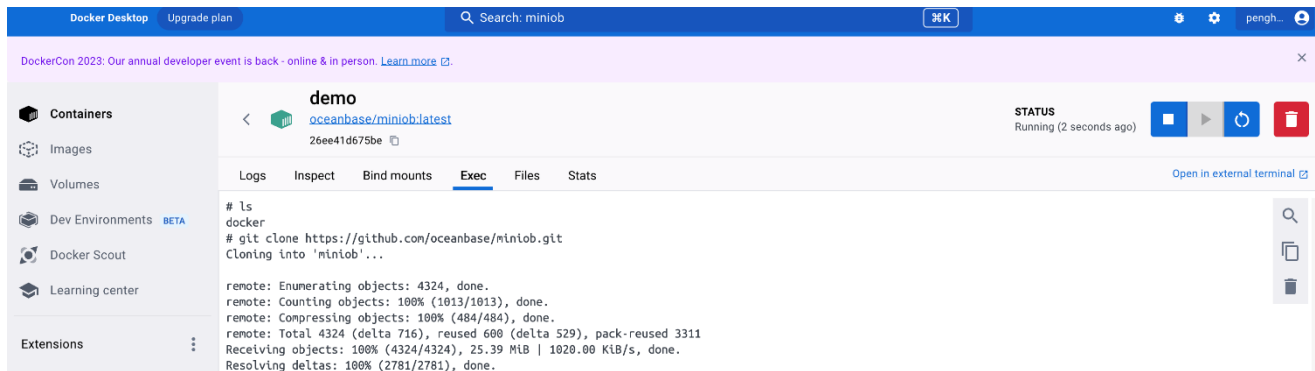
- 在Docker中直接下载官方镜像。
 - 在Docker顶部搜索框搜索miniob, oceanbase/miniob即为官方镜像



- 点击打开后, 选择pull并run即可运行

miniOB编译

- 进入容器，下载并编译miniOB。
 - 在Exec下运行 `git clone https://github.com/oceanbase/minibob.git` 从gthub上下载miniOB。



- 进入minibob目录，使用 `bash build.sh --make -j4` 进行编译

```
# cd minibob
# ls
benchmark  cmake          CODE_OF_CONDUCT.md  deps  docs  etc  NOTICE  src  tools
build.sh   CMakeLists.txt  CONTRIBUTING.md     docker  Doxyfile  License  README.md  test  unittest
# bash build.sh --make -j4
build.sh --make -j4
create soft link for build_debug, linked by directory named build
cmake -DCMAKE_EXPORT_COMPILE_COMMANDS=1 --log-level=STATUS /root/minibob -DCMAKE_BUILD_TYPE=Debug -DDEBUG=ON
-- The C compiler identification is GNU 11.4.0
-- The CXX compiler identification is GNU 11.4.0
-- Detecting C compiler ABI info
```

```
[100%] Linking CXX executable ../bin/pidfile_test
[100%] Built target pidfile_test
[100%] Linking CXX executable ../bin/ring_buffer_test
[100%] Linking CXX executable ../bin/record_manager_test
[100%] Built target persist_test
[100%] Built target ring_buffer_test
[100%] Built target record_manager_test
#
```

运行miniOB

- 运行miniOB

- 编译完成后, 进入build目录, 使用 `./bin/observer -s miniob.sock -f ../etc/observer.ini &` 在后台启动服务

```
# cd build
# ./bin/observer -s miniob.sock -f ../etc/observer.ini &
# Successfully load ../etc/observer.ini
```

- 启动服务后, 使用 `./bin/obclient -s miniob.sock` 启动miniOB, 完成

```
# ./bin/obclient -s miniob.sock
miniob > help
Commands
show tables;
desc `table name`;
create table `table name` (`column name` `column type`, ...);
create index `index name` on `table` (`column`);
insert into `table` values(`value1`,`value2`);
update `table` set column=value [where `column`=`value`];
delete from `table` [where `column`=`value`];
select [ * | `columns` ] from `table`;
miniob >
```